

Unit 7: Rewriting Polynomial Expressions

1. Factor $24r^3 + 18r^2$.

$$\begin{aligned}
 &24r^3 + 18r^2 \\
 &6(4r^3 + 3r^2) \text{ - factor out } 6 \\
 &\underline{6r^2(4r + 3)} \text{ - factor out } r^2
 \end{aligned}$$

2. Factor $6x^2 + 37x + 6$.

$$\begin{array}{c}
 \swarrow \quad \searrow \\
 36 + 1
 \end{array}$$

	x	6
$6x$	$6x^2$	$36x$
1	x	6

$$(6x + 1)(x + 6)$$

3. Factor $2x^2 - 9x - 5$

$$\begin{array}{c}
 \swarrow \quad \searrow \\
 -10 + 1
 \end{array}$$

	x	-5
$2x$	$2x^2$	$-10x$
1	x	-5

$$(2x + 1)(x - 5)$$

4. Rewrite the expression $4x^2 - 16y^2$ as a difference of two squares.

$$\begin{array}{ccc}
 \downarrow & \downarrow & \downarrow \downarrow \\
 2 & x & 4 & y
 \end{array}$$

$$(2x + 4y)(2x - 4y)$$